



LECTURE 2

GEOSPATIAL DATA: SOURCES, MODALITIES, AND APPLICATIONS

Geospatial Representation Learning

PRESS – OR SPACE. →

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Learning Outcomes

Lecture

- Understand and compare major geospatial data sources from GIS, remote sensing, meteorology, geophysics, and environmental monitoring.
- Distinguish common geospatial data modalities, including raster images, vector layers, point data, point clouds, time series, trajectories, and gridded fields.
- Explain key properties of geospatial data, including spatial resolution, temporal resolution, spectral resolution, coverage, uncertainty, and coordinate reference systems.
- Describe common geospatial data formats and access patterns used in Python-based workflows.
- Match geospatial data sources and modalities to suitable environmental and socio-ecological applications.
- Identify practical challenges such as scale mismatch, missing data, sampling bias, spatial autocorrelation, and heterogeneous data quality.

Lab

- Access and preprocess raster, vector, and time-series geospatial data using Python-based tools.
- Combine multiple geospatial data layers into a common spatial reference and resolution.
- Extract training samples or analysis regions from geospatial datasets.
- Evaluate practical limitations of data quality, coverage, scale, and interoperability.



PRACTICAL 2

WORKING WITH GEOSPATIAL DATA SOURCES

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